BEING FLAT IS FPQC LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's parenthetical '(Also see Morphisms, Lemma X for a direct proof.)' is part of the delivered proof and is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Flatness of a morphism of schemes is defined on the maps of local rings. Morphisms, Definition morphisms-definition-flat, invoked at proof lines 6987-6989. Justifies replacing the geometric statement by the algebraic one.
- Composition of flat local homomorphisms is flat, and a flat local homomorphism is faithfully flat. Algebra, Lemmas algebra-lemma-composition-flat and algebra-lemma-local-flat-ff, invoked at proof lines 6993-6998. Gives the forward direction and makes B -> C faithfully flat.
- Permanence of flatness along a faithfully flat extension. Algebra, Lemma algebra-lemma-flat-permanence, invoked at proof lines 6999-7000. Gives the converse direction.

Source reference. Original lemma 00HC: sources/stacks-chow/tags.txt, source lines 638–638. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 039V: sources/stacks-chow/tags.txt, source lines 4286–4286. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01U3: sources/stacks-chow/tags.txt, source lines 2396–2396. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02JZ: sources/stacks-chow/tags.txt, source lines 3354–3354. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is flat" is fpqc local on the source.*

Proof. Since flatness is defined in terms of the maps of local rings (Morphisms, Definition 01U3) what has to be shown is the following algebraic fact: Suppose $A \to B \to C$ are local homomorphisms of local rings, and assume $B \to C$ is flat. Then $A \to B$ is flat if and only if $A \to C$ is flat. If $A \to B$ is flat, then $A \to C$ is flat by Algebra, Lemma 00HC. Conversely, assume $A \to C$ is flat. Note that $B \to C$ is faithfully flat, see Algebra, Lemma 00HR. Hence $A \to B$ is flat by Algebra, Lemma 039V. (Also see Morphisms, Lemma 02JZ for a direct proof.) □